INTERNATIONAL UNIVERSITY OF APPLIED SCIENCES

CSEMCSPCSP01: Project Computer Science

“Comparative Analysis between Fixed-Time Traffic Signal Control System and Adaptive Traffic Signal Control System”

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1. Abstract

With the increasing number of vehicles in urban areas, traffic congestion (private and public vehicles) and pedestrian safety, especially for elderly, children and special users, have become concerning challenges for modern transportation systems. Typical traffic signal systems are generally based on fixed-time scheduling, where signals operate according to predefined durations regardless of real-time traffic conditions and designed to focus mostly on motorized vehicle movements. This can increase pedestrian waiting times, reduce accessibility, and create safety risks at busy intersections.

The proposed solution suggests design and implementation of an adaptive traffic signal system through software simulation for intersections with vehicle lanes and pedestrian crossings. The implementation is developed using Python and combines a console-based simulation with graphical visualization. The project applies core computer science concepts such as queue data structures, state-based modelling, simulation, control logic, and algorithm design. Mathematical evaluation is included to assess system performance through measurable metrics such as average waiting time, traffic throughput, and fairness ratio between pedestrians and vehicles. This project addresses this issue by designing a traffic signal system that dynamically adjusts according to the motorized traffic flow and pedestrians crossing requests.

2. Introduction and Background

2.1 Introduction

Urban transportation systems are experiencing unprecedented pressure due to rapid population growth, urbanization, and increasing vehicle ownership. As cities continue to expand, managing traffic congestion while ensuring the safety and mobility of pedestrians has become one of the most significant challenges faced by Transportation Authorities worldwide. Intersections represent critical points within road networks where conflicts between vehicles and pedestrians frequently occur, making efficient traffic signal control essential for both traffic management and public safety.

Traditional traffic signal systems generally operate using fixed timing plans that allocate predetermined green, amber, and red signal durations regardless of actual traffic conditions. While such systems are relatively simple to implement and maintain, they often fail to respond effectively to fluctuating traffic demands throughout the day. Consequently, vehicles may experience unnecessary delays during low-traffic periods, while pedestrians may be required to wait excessive amounts of time before receiving a crossing signal (Hassan et al., 2014).

Recent advances in intelligent transportation systems have led to the development of adaptive traffic signal control approaches capable of adjusting signal timings dynamically based on real-time traffic conditions. These systems utilize information collected from sensors, cameras, and other monitoring devices to optimize traffic flow and reduce congestion. Research has demonstrated that adaptive traffic signal systems can significantly improve intersection efficiency compared with traditional fixed-time approaches (Singh et al., 2024).

Despite these advancements, many adaptive traffic management systems continue to prioritize vehicular traffic flow while treating pedestrian requests as secondary considerations. Although minimizing vehicle congestion remains an important objective, excessive pedestrian waiting times can negatively affect accessibility, safety, and user satisfaction. This issue is particularly relevant for vulnerable road users such as elderly individuals, children, and persons with mobility impairments, who may encounter greater difficulty when crossing busy intersections (Akyol et al., 2024).

The growing emphasis on smart city development has increased interest in transportation solutions that balance the needs of all road users rather than focusing exclusively on vehicles. Pedestrian-centric traffic management has therefore emerged as an important area of research within intelligent transportation systems. Such approaches seek to improve pedestrian mobility while maintaining acceptable traffic performance through the use of intelligent decision-making algorithms and real-time data analysis.

This project contributes to this research area by proposing a pedestrian-centric adaptive traffic signal control system that incorporates pedestrian requests, queue lengths, and waiting times into the signal decision-making process. The proposed solution utilizes queue-based priority scheduling to dynamically allocate signal phases based on current intersection conditions. Through simulation-based evaluation, the project aims to investigate whether pedestrian waiting times can be reduced without significantly compromising vehicular traffic efficiency.

2.2 Background

Traffic congestion has become a persistent issue in modern cities due to increasing travel demand and limited roadway capacity. According to intelligent transportation research, traffic delays contribute not only to economic losses but also to increased fuel consumption, environmental pollution, and reduced quality of life for urban residents (Shams et al., 2023). As a result, transportation engineers continuously seek innovative methods to improve the efficiency of signalized intersections.

Adaptive Traffic Signal Control (ATSC) systems represent a significant advancement in traffic management technology. These systems continuously monitor traffic conditions and adjust signal timings accordingly. Various adaptive approaches have been proposed in the literature, including rule-based systems, fuzzy logic controllers, optimization-based methods, and machine learning techniques (Singh et al., 2024). Several studies have reported substantial reductions in traffic delay through the use of adaptive control mechanisms (Gao et al., 2017; Muresan et al., 2019).

Recent research has emphasized the importance of integrating pedestrian demand directly into signal control algorithms. Akyol et al. (2024) demonstrated that pedestrian delay can be reduced through multi-objective optimization techniques that balance pedestrian and vehicular requirements. Similarly, Gholami et al. (2024) highlighted the importance of considering pedestrian timing requirements when coordinating signalized intersections.

This research proposes a computationally efficient traffic signal control strategy based on queue management and dynamic priority scheduling. By incorporating pedestrian waiting times and queue lengths directly into signal decision-making processes, the proposed system seeks to achieve a more balanced and equitable allocation of intersection resources.

2.3 Problem Statement

The problem addressed by this research is the lack of a simple, efficient, and pedestrian-focused adaptive traffic signal control strategy capable of dynamically balancing pedestrian requests and vehicular traffic conditions. This study seeks to investigate whether a queue-based priority scheduling approach can reduce pedestrian waiting times while maintaining acceptable levels of vehicle traffic performance within a simulated urban intersection environment.

3. Objectives

3.1 Primary Objective

The primary objective of this research is to develop and simulate an adaptive pedestrian-centric traffic signal control system that reduces pedestrian waiting times while maintaining efficient vehicular traffic movement at urban intersections.

3.2 Secondary Objectives

To achieve the primary objective, the following specific objectives have been identified:

a. The effectiveness of the proposed system is measured using performance metrics such as average pedestrian waiting time, vehicle delay, throughput, and fairness.
b. The performance of the adaptive traffic signal controller is compared against a conventional fixed-time traffic signal strategy.

3.3 Research Questions

The research is guided by the following questions:

a. Can an adaptive pedestrian-centric traffic signal system significantly reduce pedestrian waiting times compared with traditional fixed-time signal control?
b. Can a balanced allocation of signal priority improve overall intersection fairness for both pedestrians and vehicles?

4. Methodology

4.1 Research Methodology

This study adopts a quantitative simulation-based research methodology to investigate the effectiveness of a pedestrian-centric adaptive traffic signal control system. Simulation is widely used within transportation research because it allows researchers to evaluate traffic management strategies under controlled conditions without affecting real-world traffic operations (Singh et al., 2024). The methodological framework consists of five major phases: system design, algorithm development, simulation implementation, performance evaluation, and results analysis.

4.2 System Development Approach

The proposed system follows an adaptive control approach in which signal timing decisions are continuously updated according to current traffic conditions. Rather than relying on fixed signal cycles, the controller dynamically determines which traffic movement should receive priority based on calculated priority scores. This approach enables the traffic controller to respond more effectively to fluctuations in demand throughout the simulation period.

4.3 Simulation Environment

The simulation environment will be developed using python. Several required libraries such as NumPy, Matplotlib, random, etc. will be imported to support development process. Python is flexible, robust and easy to implement, making it the preferred choice for implementation process. The simulation environment will model a four-way urban intersection containing vehicle lanes and pedestrian crossings. Traffic entities will arrive according to predefined arrival rates and will interact with the adaptive signal controller throughout the simulation period.

4.4 Traffic Modelling

The simulation will represent two primary categories of road users: vehicles and pedestrians. Vehicle arrivals will be generated at specified rates to simulate traffic conditions. Similarly, pedestrian requests will be generated to represent crossing demand. Each vehicle and pedestrian entering the simulation will be assigned attributes such as arrival time, queue position, and waiting duration. These attributes will be continuously updated as the simulation progresses. The use of queue structures allows the system to maintain accurate representations of traffic conditions while supporting efficient decision-making processes.

4.5 Proposed Adaptive Control Algorithm

Adaptive traffic control systems continuously monitor traffic conditions and adjust signal timings in response to changing demand (Singh et al., 2024). Building upon this concept, the proposed algorithm introduces a pedestrian-centric decision-making mechanism. It is a queue-based priority scheduling algorithm. The proposed algorithm explicitly incorporates pedestrian demand into the decision-making process. During each signal cycle, the controller evaluates three key variables: pedestrian queue length, pedestrian waiting time and vehicle queue length. These variables are combined to produce a dynamic priority score that determines which movement should receive the next green signal phase.

The proposed priority calculation is expressed as:

Priority Score = (Pedestrian Queue × W_1) × (Pedestrian Waiting Time × W_2) − (Vehicle Queue × W_3)

Where,

W_1 represents the importance assigned to pedestrian queue length,

W_2 represents the importance assigned to pedestrian waiting time,

W_3 represents the importance assigned to vehicle congestion.

A higher priority score indicates a greater need for pedestrian service, while lower scores favour continued vehicle movement. Additional constraints are incorporated to prevent starvation, ensuring that neither pedestrians nor vehicles experience excessive delays. This fairness-oriented approach is consistent with recommendations found in pedestrian-aware traffic management research (Akyol et al., 2024).

4.6 Signal Decision Process

The signal decision process operates continuously throughout the simulation. First, traffic and pedestrian data are collected from the simulation environment. The queue management module then updates all waiting times and queue lengths. Subsequently, the adaptive control algorithm calculates priority scores and evaluates whether signal changes are required. If pedestrian demand exceeds predefined thresholds, the controller may allocate additional crossing opportunities. Conversely, if vehicle congestion becomes excessive, the algorithm may temporarily extend vehicle green phases. The resulting signal decision is then implemented, and the process repeats until the simulation concludes. Figure 1 explains the signal decision process.

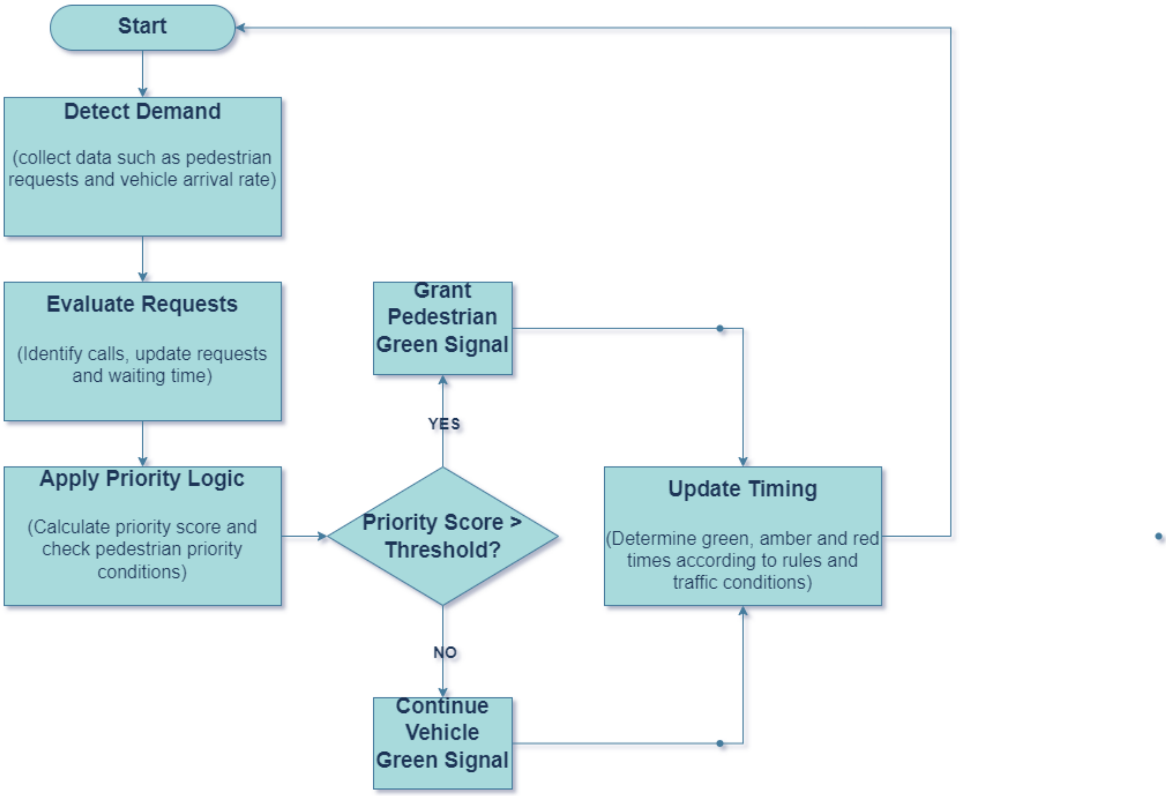

Figure 1: Adaptive Signal Decision Process Flow Chart

(Source: Created by author)

4.7 Performance Evaluation

The effectiveness of the proposed traffic signal controller will be assessed using several quantitative performance indicators.

- Average Pedestrian Waiting Time (APWT) measures the average duration pedestrians wait before receiving a crossing signal.
- Average Vehicle Delay (AVD) measures the average time vehicles spend waiting at the intersection.
- Throughput measures the number of vehicles successfully passing through the intersection during the simulation period.
- Fairness Index evaluates how equitably signal priority is distributed between pedestrians and vehicles.
- Signal Response Time measures the speed at which the controller responds to pedestrian requests.

Together, these metrics provide a comprehensive assessment of both operational efficiency and pedestrian accessibility.

4.8 Experimental Setup

To evaluate the effectiveness of the proposed adaptive traffic signal controller, a simulation experiment will be conducted. Testing traffic scenarios is a common practice in transportation research because it enables the robustness of a traffic control strategy to be evaluated under different operational environments (Gao et al., 2017). The simulation environment will model a four-way signalized intersection containing both vehicle lanes and pedestrian crossings.

Scenario will be tested using conventional fixed time signal control and adaptive signal control. Performance metrics collected from both systems will be compared to determine the effectiveness of the proposed approach. The comparison will focus on average pedestrian waiting time, average vehicle delay, throughput, queue length, fairness index and signal response time.

4.9 Assumptions and Constraints

Assumptions:

- All pedestrians obey traffic signals.
- Vehicle arrivals follow predefined traffic patterns.
- Sensors operate without failure in the simulation.
- Only one intersection is considered.

Constraints:

- Real-world weather conditions are not modelled.
- Driver behaviour variability is simplified.
- Emergency vehicle prioritization is excluded.
- Multi-intersection coordination is outside the project scope.

5. Timeline and Milestones

A structured development plan is essential to ensure that the project progresses systematically and that all research objectives are achieved within the available timeframe. The project will be divided into several phases, beginning with literature review and system design, followed by implementation, testing, evaluation, and final documentation. The timeline has been designed to support an iterative development process in which each phase builds upon the outcomes of the previous stage. This approach reduces development risks and allows continuous validation of the proposed traffic signal control strategy. The following table describes the time period required during each phase.

5.1 Project Schedule

Table 1: Project Schedule.

Phase	Activity	Duration
Phase 1	Literature Review and Problem Analysis	Week 1 - 2
Phase 2	System Design and Architecture Development	Week 3
Phase 3	Algorithm Development	Week 4 - 5
Phase 4	Python Simulation Implementation	Week 6 - 7
Phase 5	Testing and Validation	Week 8
Phase 6	Performance Evaluation and Analysis	Week 9
Phase 7	Final Report	Week 10

(Source: Created by author)

5.2 Key Milestones

Several milestones have been established to measure project progress and ensure successful completion.

- The first milestone involves completing the literature review and identifying gaps in existing pedestrian-aware traffic management systems. This milestone provides the theoretical foundation for the proposed solution.
- The second milestone focuses on designing the adaptive signal control architecture and developing the queue-based priority scheduling algorithm. At this stage, the core functionality of the system will be formally specified.
- The third milestone involves implementing the simulation environment in Python and integrating all system modules. Successful completion of this milestone will result in a functional prototype capable of simulating intersection behaviour.
- The fourth milestone consists of conducting simulation experiments under different traffic scenarios and collecting performance data. This phase will generate the results required for evaluating the effectiveness of the proposed system.
- The final milestone involves analysing the results, discussing findings, and preparing the complete project report.

6. System Design and Architecture

The architecture follows a modular design approach that separates data collection, queue management, decision-making, and signal control functions. This modular structure improves maintainability, scalability, and future extensibility of the system.

The system consists of four major layers. Each layer performs a specific function within the overall traffic management process.

Data Collection Layer:

The data collection layer is responsible for gathering information about vehicles and pedestrians approaching the intersection. In a real-world deployment, data may be obtained from induction loop detectors, cameras, infrared sensors, or pedestrian push-button systems (Singh et al., 2024). Within the simulation environment, traffic and pedestrian arrivals will be generated programmatically to represent realistic intersection conditions.

The collected information includes:

- Vehicle arrival rates
- Vehicle queue lengths
- Pedestrian crossing requests
- Pedestrian waiting times

This information serves as the primary input for subsequent decision-making processes.

Queue Management Layer:

The queue management layer organizes incoming traffic data and maintains records of waiting vehicles and pedestrians. Queues are widely used in transportation systems to represent traffic accumulation and waiting behaviour (Shams et al., 2023). In the proposed system, separate queue structures are maintained for vehicles and pedestrians. Each queue entry contains information such as:

- Arrival time
- Waiting duration
- Queue position

The queue management module continuously updates these values as the simulation progresses.

Decision Engine:

This component receives data from the queue management layer and applies the adaptive priority scheduling algorithm developed in this research. The algorithm evaluates current traffic conditions and calculates a priority score that determines whether signal priority should be allocated to pedestrians or vehicles.

By considering both waiting times and queue lengths, the decision engine seeks to achieve a balanced allocation of intersection resources.

Signal Control Layer:

The signal control layer is responsible for implementing decisions generated by the decision engine. This layer controls:

- Vehicle traffic lights
- Pedestrian crossing signals
- Signal phase transitions

Once a signal decision has been made, the controller updates signal states and communicates the new phase to all road users within the simulation environment.

The cycle then repeats continuously until the simulation concludes.

6.1 Architectural Workflow

The operational workflow of the proposed system consists of the following sequence:

- Traffic and pedestrian data are collected.
- Queue lengths and waiting times are updated.
- The decision engine calculates priority scores.
- Signal allocation decisions are generated.
- Traffic signals are updated.
- System performance metrics are recorded.
- The process repeats for the next simulation cycle.

This continuous feedback loop enables adaptive traffic management and ensures that signal decisions remain responsive to changing conditions.

6.2 UML Use Case Design

The UML Use Case Diagram illustrates interactions between system actors and the traffic signal controller. Three primary actors participate within the proposed system:

a. Pedestrian:

The pedestrian actor initiates crossing requests and interacts with the traffic signal system through a crossing request mechanism.

b. Vehicle Detection Sensor:

The vehicle detection sensor continuously monitors traffic conditions and provides information regarding vehicle arrivals and queue lengths.

c. Traffic Controller

The traffic controller serves as the central decision-making component responsible for processing traffic information and allocating signal phases.

The use case model provides a high-level representation of system functionality and supports software design activities.

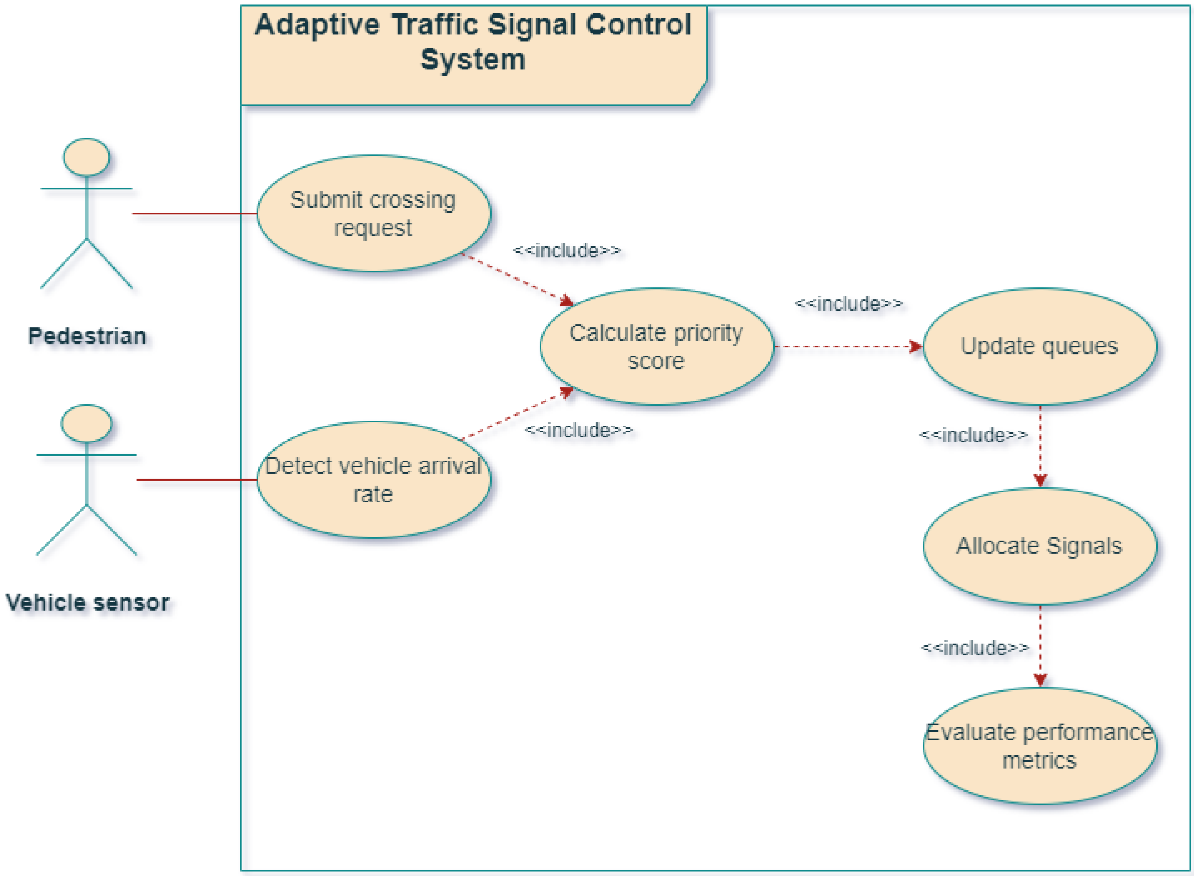

Figure 2: UML Use Case Diagram of Adaptive Traffic Signal Control System

(Source: Created by Author)

6.3 UML Class Diagram

The system is composed of five primary classes:

a. Vehicle
b. Pedestrian
c. QueueManager
d. SignalController
e. StatisticsCollector

Each class is responsible for a specific aspect of the simulation environment.

The Vehicle and Pedestrian classes represent traffic entities entering the system. These entities are managed by the QueueManager, which maintains waiting queues and updates waiting times throughout the simulation.

The SignalController serves as the core decision-making component and interacts with the QueueManager to retrieve current traffic conditions before allocating signal phases.

Finally, the StatisticsCollector gathers performance metrics throughout the simulation and stores the information for later analysis.

This modular design supports future system enhancements and simplifies testing and maintenance activities.

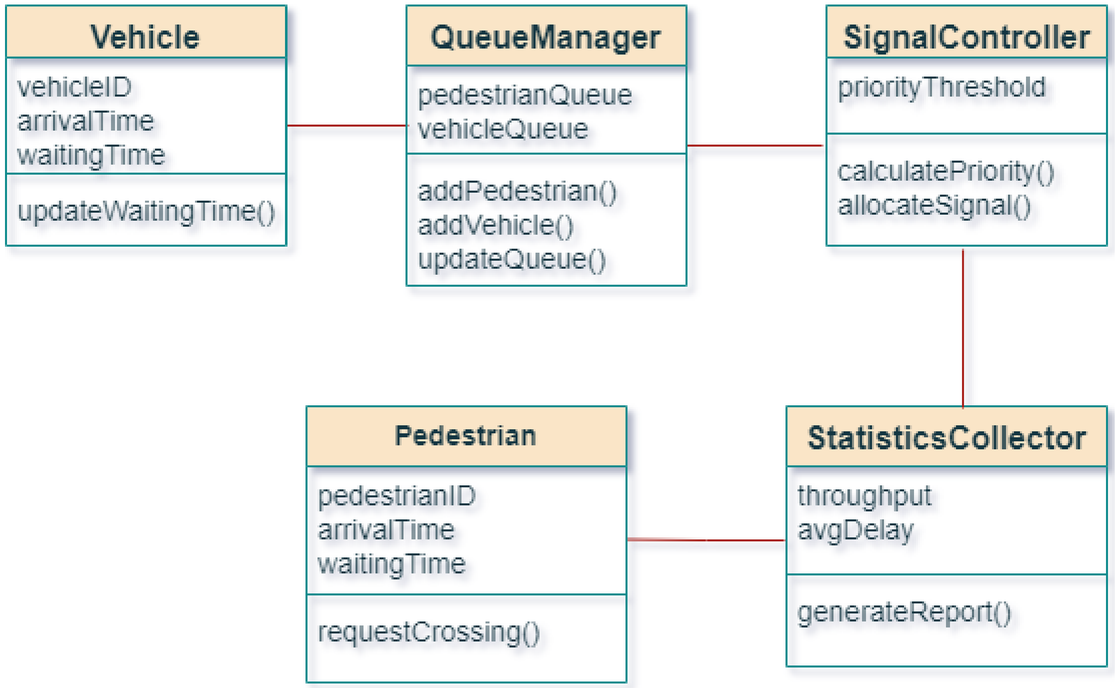

Figure 3: UML Class Diagram of Adaptive Traffic Signal Control System

(Source: Created by author)

6.4 Performance Evaluation Metrics

The effectiveness of any traffic signal control strategy must be evaluated using objective and quantifiable performance indicators. Several studies evaluating adaptive traffic signal control systems have utilized metrics such as delay, throughput, queue length, and waiting time to measure operational effectiveness (Gao et al., 2017; Muresan et al., 2019). Consistent with established transportation research practices, this project adopts a combination of pedestrian-focused and vehicle-focused performance indicators.

The selected metrics allow the proposed adaptive controller to be compared with a conventional fixed-time traffic signal system under identical simulation conditions.

6.4.1 Average Pedestrian Waiting Time

Average Pedestrian Waiting Time (APWT) is one of the most important evaluation metrics in this study because pedestrian accessibility represents the primary focus of the proposed system (Harely Amado et al., 2020). This metric measures the average amount of time that pedestrians must wait before receiving a crossing signal. Long waiting times can reduce pedestrian satisfaction and may encourage unsafe crossing behaviour, particularly in busy urban environments (Hassan et al., 2014).

The metric is calculated as follows:

$$\text{APWT} = \frac{\Sigma Pedestrian\ Waiting\ Time}{Number\ of\ Pedestrians}$$

Lower APWT values indicate improved pedestrian service and greater system responsiveness.

6.4.2 Average Vehicle Delay

Average Vehicle Delay (AVD) measures the average amount of time vehicles spend waiting at the intersection before being allowed to proceed.

The Average Vehicle Delay is calculated using the following formula:

$$\text{AVD} = \frac{\Sigma Vehicle\ Delay}{Number\ of\ Vehicles}$$

Lower values indicate better traffic flow performance.

6.4.3 Throughput

Throughput represents the total number of vehicles that successfully pass through the intersection during the simulation period. Throughput is widely used in transportation engineering as a measure of intersection efficiency because it reflects the ability of the traffic control system to process traffic demand effectively (Shams et al., 2023).

The throughput calculation is expressed as:

$$\text{Throughput} = \frac{Number\ of\ Vehicles\ Passed}{Simulation\ Duration}$$

Higher throughput values indicate greater operational efficiency.

6.4.4 Queue Length

Queue length measures the number of vehicles or pedestrians waiting at any given time. Queue accumulation is often associated with traffic congestion and reduced mobility. Monitoring queue lengths provides additional insight into the effectiveness of the proposed scheduling algorithm. Average queue length will be recorded throughout the simulation and compared with the results obtained from a fixed-time signal control strategy. Smaller queue lengths generally indicate more efficient traffic management.

6.4.5 Fairness Index

A system that completely prioritizes vehicles may significantly disadvantage pedestrians, whereas excessive pedestrian prioritization may generate unacceptable traffic congestion. Therefore, fairness must be considered alongside efficiency.

The Fairness Index (FI) is defined as:

$$FI = \frac{Minimum\ Service\ Level}{Maximum\ Service\ Level}$$

Values closer to 1 indicate a balanced allocation of signal priority between pedestrians and vehicles.

The fairness metric is particularly relevant for pedestrian-centric systems because it evaluates whether improvements in pedestrian service are achieved without imposing disproportionate penalties on vehicular traffic.

6.4.6 Signal Response Time

Signal Response Time measures the duration between a pedestrian crossing request and the system's decision to allocate a crossing opportunity. Adaptive traffic signal systems are expected to respond more quickly than traditional fixed-time systems because decisions are based on real-time traffic conditions (Singh et al., 2024). Lower response times indicate a more adaptive and responsive traffic management strategy.

6.4.7 Comparative Evaluation

The proposed adaptive traffic signal controller will be compared against a traditional fixed-time signal control system. Both systems will be tested under identical traffic conditions to ensure a fair comparison. The resulting performance data will be presented using tables, charts, and statistical summaries to facilitate detailed analysis. By examining multiple performance indicators simultaneously, the study aims to provide a balanced assessment of the strengths and limitations of the proposed adaptive control strategy.

6.5 Expected Results

The primary expected outcome is a reduction in pedestrian waiting time due to the incorporation of pedestrian demand directly into signal allocation decisions. The adaptive controller is also expected to demonstrate improved responsiveness to changing traffic conditions because signal phases are determined dynamically rather than through predetermined schedules. Although some increase in vehicle delay may occur during periods of high pedestrian demand, the use of balanced priority scheduling is expected to prevent excessive traffic congestion. Furthermore, the fairness index is expected to improve because the controller considers both pedestrian and vehicular requirements during decision-making. Overall, the proposed system is expected to demonstrate that pedestrian accessibility can be improved without substantially reducing intersection efficiency.

Table 2 provides an idea of the expected results based on this scientific reasoning for the performance improvements.

Table 2: Expected comparison between fixed-time and adaptive traffic signal control systems

Metric	Traditional System Performance	Adaptive System Performance
Pedestrian Waiting Time	High	Lower
Vehicle Delay	Moderate	Similar or Slightly Higher
Throughput	Moderate	Higher
Queue Length	Higher	Lower
Fairness	Lower	Higher
Response Time	Slow	Faster

(Source: Created by author)

6.6 Data Visualization

The proposed system is expected to demonstrate improvements in several performance indicators, particularly pedestrian waiting time and signal responsiveness. The collected simulation data will be analysed to determine whether the proposed adaptive traffic signal controller achieves its objectives. Several visualization techniques will be used to support result interpretation.

The evaluation will be performed using both numerical and graphical methods. Numerical analysis will include average values, percentage improvements, and comparative summaries. Graphical analysis will include bar charts, line graphs, and trend visualizations generated using Python. The combination of numerical and graphical evaluation provides a comprehensive assessment of system performance and facilitates interpretation of experimental findings.

Comparative line graphs will be used to visualize pedestrian waiting time, vehicle delay, throughput analysis, queue length and fairness index. This analysis provides insight into congestion management performance and evaluate the balance between pedestrian and vehicular service quality. Fairness analysis is particularly important because the objective of the proposed system is not to maximize pedestrian priority at all costs, but rather to achieve an equitable allocation of signal resources.

In addition to graphical analysis, summary tables will be generated for experimental scenario. These tables will facilitate direct comparison between the adaptive and fixed-time control strategies.

7. Implementation and Results

7.1 Development Environment and Source Code Structure

The simulation model is developed using Python due to its extensive support for scientific computing, data analysis, and visualization. Several software libraries have been utilized during development. NumPy supports mathematical calculations, Pandas assist with data management and analysis. The Random library is used to generate vehicle and pedestrian arrivals during simulation runs. The Time library assists in simulation timing and event management. Matplotlib is used to generate performance graphs. The software implementation follows a modular architecture in which each component performs a specific task.

- main.py: Acts as the entry point of the simulation and coordinates interactions among all system components.
- vehicle.py: Defines the Vehicle class and manages vehicle-related attributes and operations.
- pedestrian.py: Defines the pedestrian class and manages pedestrian requests and waiting times.
- queue_manager.py: Maintains vehicle and pedestrian queues and updates waiting times throughout the simulation.
- signal_controller.py: Implements the adaptive scheduling algorithm and controls signal allocation decisions.
- statistics.py: Collects and processes performance metrics generated during simulation runs.
- config.py: Stores configuration settings including traffic rates, signal timing parameters, and simulation duration.

This modular design improves maintainability and facilitates future extensions of the system.

7.2 Simulation Workflow

The simulation process follows a sequence of steps that model intersection behaviour. Initially, vehicles and pedestrians are generated according to predefined arrival rates. The queue manager then stores arriving entities and updates their waiting times during each simulation cycle. The signal controller continuously evaluates traffic conditions and calculates a priority score based on queue lengths and waiting times. Based on the calculated priority score, the controller allocates signal phases and updates the state of the intersection. Throughout the simulation, performance metrics are recorded for later analysis. The process continues until the simulation duration has been completed.

7.3 Results and Analysis

At the conclusion of each simulation run, performance metrics are summarized and presented through graphs and statistical tables. The results are compared against a conventional fixed-time traffic signal controller to evaluate the effectiveness of the proposed adaptive approach. The analysis focuses on determining whether pedestrian waiting times can be reduced while maintaining acceptable levels of vehicular traffic performance.

A combination of bar charts is used to compare average pedestrian waiting times and vehicle delay under fixed-time and adaptive traffic control systems.

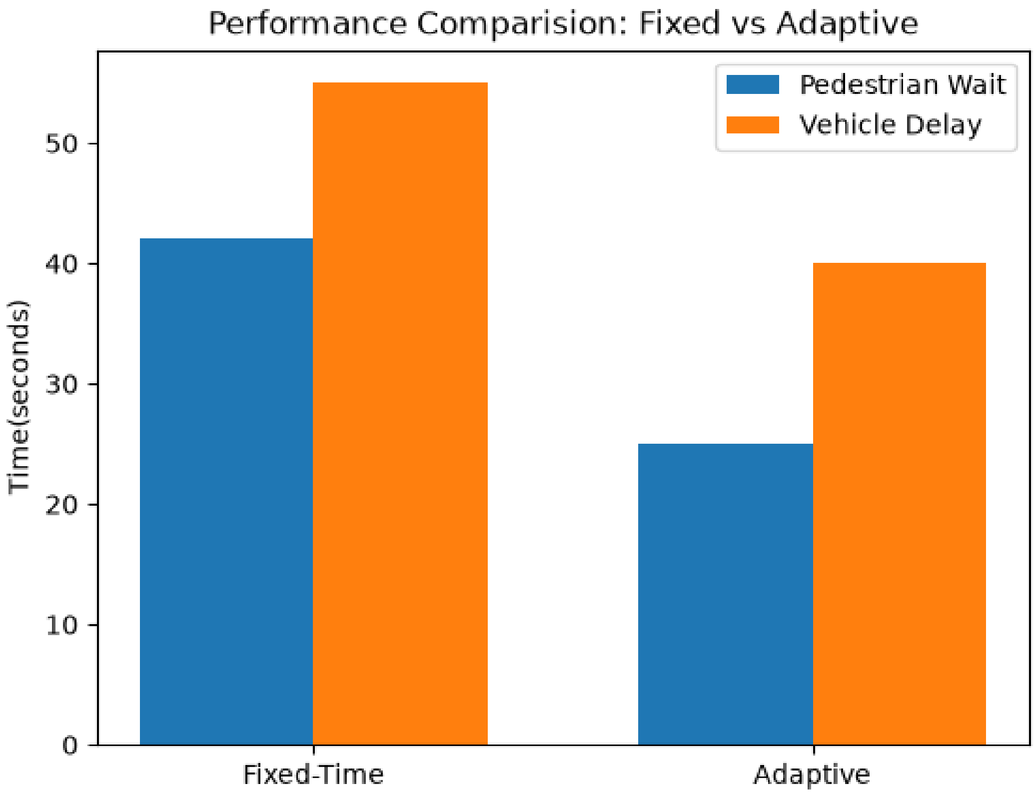

Figure 4: Comparing Average Pedestrian Waiting Time and Average Vehicle Delay under Fixed-Time VS. Adaptive Signal Controller

(Source: Created by author)

As shown in figure 4, the reduction in waiting time indicates improved pedestrian accessibility and demonstrates the effectiveness of the algorithm. This visualization illustrates that improvements in pedestrian service are achieved without introducing excessive traffic congestion.

Comparative line graphs are used to visualize pedestrian waiting time, vehicle delay, throughput analysis, queue length and fairness index. This analysis provides insight into congestion management performance and evaluate the balance between pedestrian and vehicular service quality.

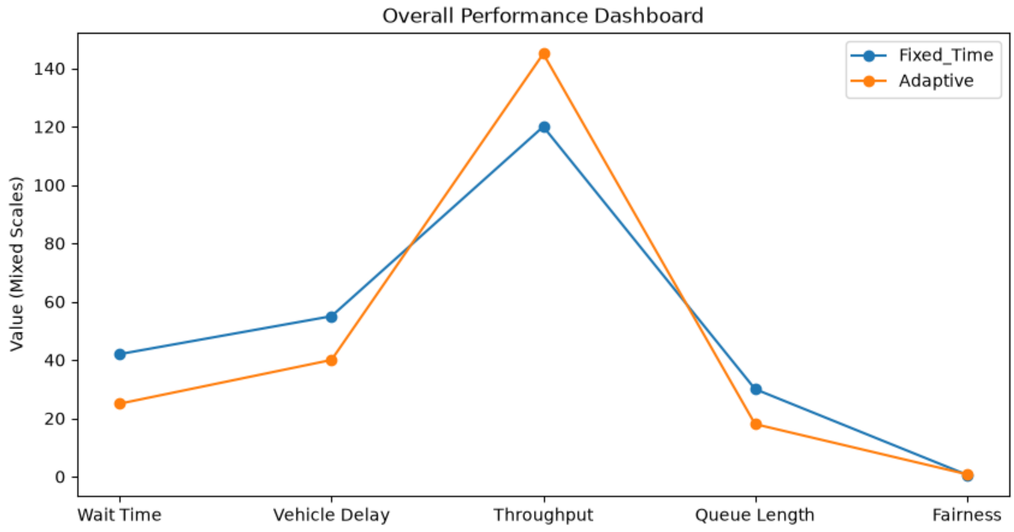

Figure 5: Overall Performance Evaluation

(Source: Created by author)

As illustrated in figure 5, there is a significant amount of reduction in pedestrian waiting time, vehicle delay and queue length as compared to conventional fixed time signal controller. The graph depicts that the adaptive signal controller shows the increase in throughput while maintaining the fairness index. Fairness analysis is particularly important because the objective of the proposed system is not to maximize pedestrian priority at all costs, but rather to achieve an equitable allocation of signal resources.

The summary table facilitate direct comparison between the adaptive and fixed-time control strategies.

Table 3: Performance Metrics Summary

Metric	Fixed-Time	Adaptive
Pedestrian Wait (s)	42	25
Vehicle Delay (s)	55	40
Throughput	120	145
Queue Length	30	18
Fairness Index	0.62	0.81

(Source: Created by author)

According to table 3, adaptive strategy illustrates the reduction in pedestrian waiting time to approximately half of the fixed time systems. Vehicle delay and queue length amounts have also noticeably plummeted using adaptive signal control algorithm. Traditional systems prioritise vehicle flow. However, adaptive system describes more throughput and better fairness index than the fixed time system.

8. Expected Challenges and Proposed Solutions

Although adaptive traffic signal control systems offer significant potential for improving transportation efficiency and pedestrian accessibility, their implementation presents several technical and operational challenges. These challenges must be identified and addressed during the design stage to ensure the reliability and effectiveness of the proposed system. This chapter discusses the primary challenges expected during the development and evaluation of the pedestrian-centric adaptive traffic signal control system. For each challenge, appropriate mitigation strategies are proposed to reduce potential risks and improve system robustness.

8.1 Real-Time Decision-Making Constraints

One of the primary challenges associated with adaptive traffic management systems is the need for real-time decision-making. Previous studies have highlighted that adaptive signal control systems must balance computational complexity with responsiveness to ensure practical deployment (Shams et al., 2023). Highly complex optimization algorithms may produce accurate decisions but can introduce delays that reduce overall system effectiveness. For this reason, the proposed research adopts a queue-based priority scheduling algorithm rather than computationally intensive machine learning approaches. The use of lightweight calculations allows signal decisions to be generated quickly while still adapting to changing traffic conditions. By maintaining a relatively simple decision-making process, the proposed system is expected to satisfy real-time operational requirements within the simulation environment.

8.2 Sensor Reliability and Data Accuracy

However, sensor failures, environmental conditions, and communication errors can affect data quality. Incorrect traffic measurements may lead to inappropriate signal decisions, reducing system performance and potentially creating safety concerns. Within the simulation environment, sensor failures will not directly occur because traffic data are generated programmatically. Nevertheless, it is important to consider this issue when evaluating the practical applicability of the proposed system. A potential mitigation strategy involves implementing fall-back control mechanisms. If traffic data become unavailable, the controller could temporarily revert to a conventional fixed-time signal plan until reliable information becomes available again. Such fail-safe mechanisms are commonly incorporated into intelligent transportation systems to improve operational reliability.

8.3 Balancing Pedestrian and Vehicle Priorities

The primary objective of this project is to improve pedestrian accessibility. However, excessive prioritization of pedestrian requests may significantly increase vehicle delays and reduce intersection throughput. This challenge reflects a fundamental trade-off frequently discussed within transportation research (Akyol et al., 2024). Transportation planners must balance competing demands to ensure equitable service for all road users. To address this challenge, the proposed algorithm incorporates both pedestrian and vehicular factors into the priority score calculation. Rather than granting absolute priority to pedestrians, the controller evaluates queue lengths and waiting times for both groups before making a decision. This balanced approach is expected to improve pedestrian service while maintaining acceptable traffic flow conditions.

8.4 Scalability to Larger Traffic Networks

The simulation developed in this research focuses on a single signalized intersection. While this approach is appropriate for evaluating the proposed algorithm, real-world transportation networks often consist of multiple interconnected intersections. Research has shown that traffic conditions at one intersection can influence neighbouring intersections, creating additional complexity for adaptive signal control systems (Singh et al., 2024). Consequently, an algorithm that performs well at a single intersection may require modification before being deployed across a larger traffic network. Future extensions of this research could investigate distributed control architectures capable of coordinating multiple intersections simultaneously. Such an approach would improve scalability and support smart city traffic management initiatives.

8.5 Simulation Accuracy

Factors such as driver behaviour, pedestrian decision-making, weather conditions, and unexpected incidents may not be fully represented within the simulation model (Muresan et al., 2019). As a result, simulation findings should be interpreted as estimates of system performance rather than exact predictions of real-world outcomes.

9. Conclusion

The research is motivated by the limitations of existing traffic control approaches, many of which prioritize vehicular efficiency while providing limited consideration for pedestrian needs. Through the integration of adaptive scheduling principles and real-time traffic monitoring concepts, the proposed system seeks to achieve a more balanced allocation of intersection resources. A simulation-based methodology had been selected to evaluate the effectiveness of the proposed solution. The simulation environment is developed using Python and model interactions between vehicles, pedestrians, and traffic signal controllers. Performance is assessed using metrics such as average pedestrian waiting time, vehicle delay, throughput, fairness, and signal response time. These metrics allow the proposed adaptive system to be compared directly with conventional fixed-time traffic signal strategies.

The project demonstrates that pedestrian waiting times can be reduced without causing substantial degradation in vehicular traffic performance. Furthermore, the research shows that relatively simple queue-based scheduling mechanisms can provide meaningful improvements in intersection management without requiring complex artificial intelligence models or expensive infrastructure investments. Beyond its immediate application to traffic signal control, the proposed research contributes to broader discussions surrounding intelligent transportation systems and smart city development. The findings may provide a foundation for future research involving multi-intersection coordination, advanced sensing technologies, machine learning integration, and large-scale urban traffic optimization.

To summarize, this project seeks to demonstrate that pedestrian-centric traffic management can improve accessibility, safety, and overall intersection performance while maintaining an appropriate balance between pedestrian and vehicular demands. The successful implementation of the proposed system would represent a step toward more equitable and efficient urban transportation infrastructure.

10. Future Prospects

Although the proposed system focuses on a single signalized intersection, several opportunities exist for future development. One possible extension involves coordinating multiple intersections within a larger urban traffic network. Such an approach would enable traffic conditions at neighbouring intersections to influence signal allocation decisions and improve overall traffic flow.

Another potential enhancement involves incorporating machine learning techniques such as reinforcement learning. Intelligent learning algorithms could enable the system to adapt automatically to changing traffic patterns without relying on predefined weighting parameters.

Future research may also integrate computer vision technologies capable of detecting pedestrians and vehicles through real-time video analysis. Additional studies could investigate the effects of weather conditions, emergency vehicle prioritization, and public transportation scheduling on traffic signal control strategies. These extensions would increase system realism and further contribute to the development of intelligent transportation systems and smart city infrastructure.

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